Section A

Answer **all** the questions in this section.

1 Enzyme **X** digests a sugar. Fig. 1.1 shows the rate of reaction of enzyme **X** and a mixture of enzyme **X** and substance **Y** on the sugar. Both reactions were carried out in a conical flask.

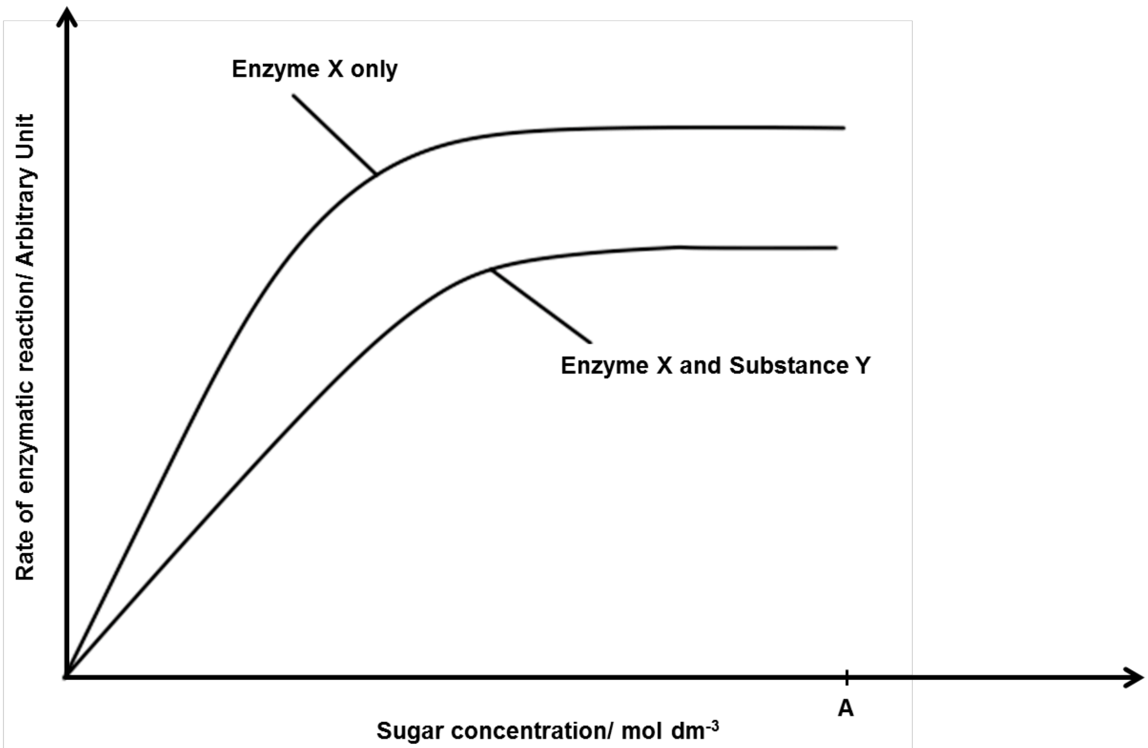

Fig. 1.1

(a) With reference to Fig. 1.1, describe and explain the effect of substance **Y** on the enzymatic reaction. [3]

(b) Substance **Y** was found in one of the products of the reaction catalysed by enzyme **X** after the substrate concentration, **A**, in Fig.1.1**.**

Predict how the trend of the **graph for enzyme X only** would continue after sugar concentrations greater than A. Sketch the part of the graph in Fig. 1.1. [1]

Enzyme **X** may be immobilised in alginate beads. Fig. 1.2 shows a comparison between the activity of free enzyme **X** and immobilised enzyme **X** over a range of temperatures. Equal concentrations of free enzyme **X** and immobilised enzyme **X** were used.

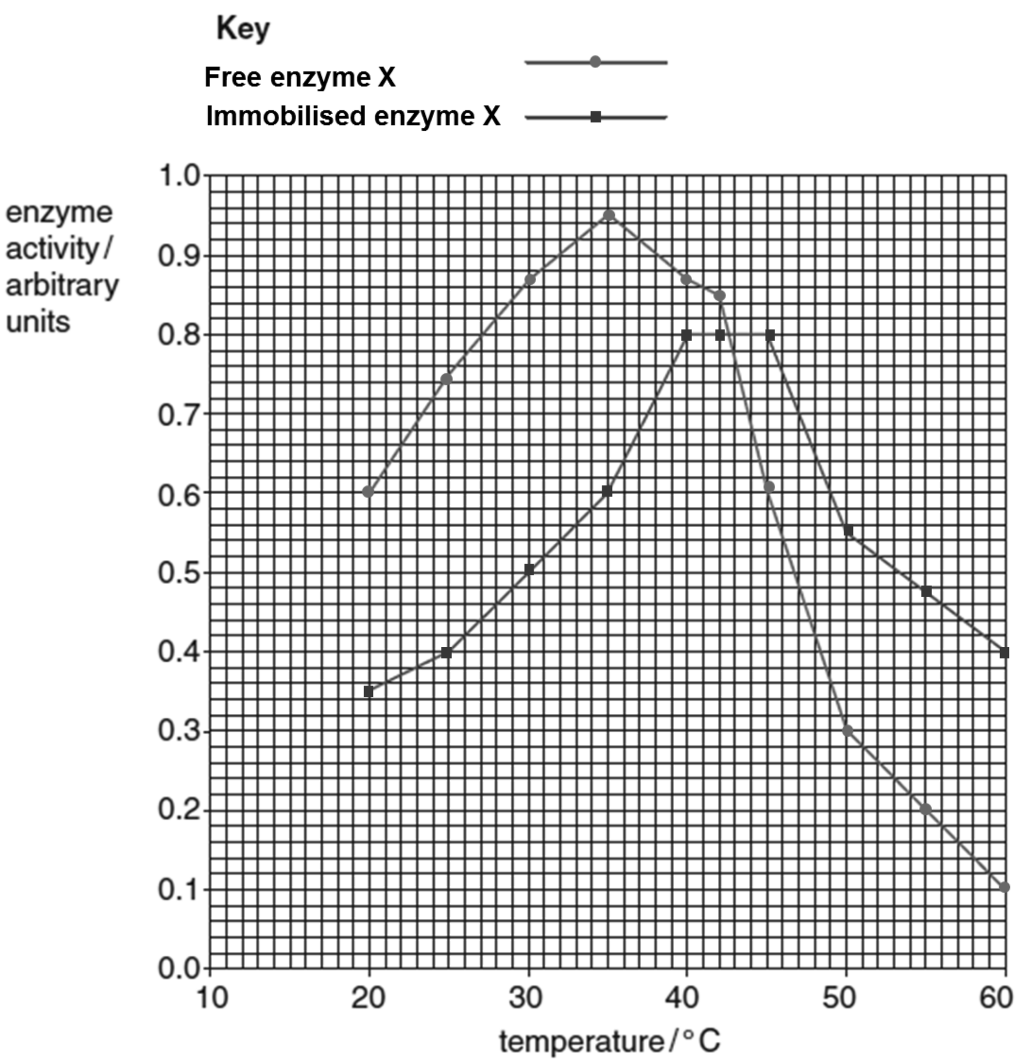

Fig. 1.2

(c) With reference to the Fig. 1.2,

(i) describe and explain the effect of temperature on the activity of free enzyme **X**. [3]

(ii) describe and explain the differences between the activity of free enzyme **X** and immobilised enzyme **X** up to 40 °C. [2]

The oil stored in the seeds of the castor oil plant *Ricinus communis* has been used in the production of pharmaceutical products such as zinc and castor oil cream, and as a lubricant. However, *Ricinus* seeds also produce a poisonous protein ricin, which is classified as an agent of bioterrorism.

Fig. 1.3 shows the structure of ricin which consists of two polypeptides – the A chain of ricin (RTA, ricin toxin A) joined to a B chain (RTB, ricin toxin B).

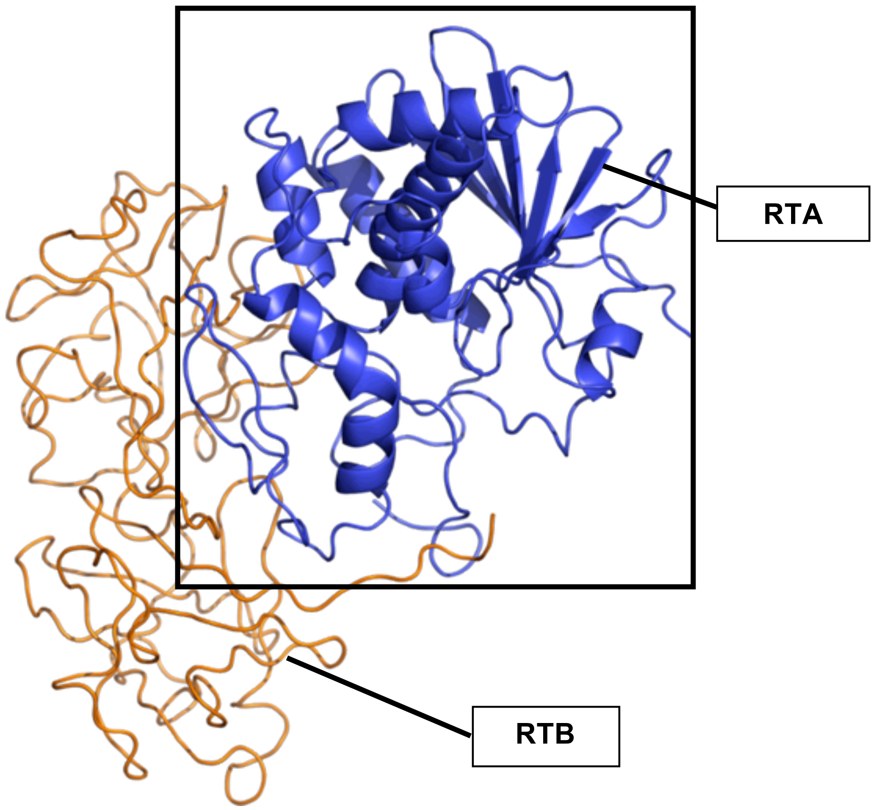

Fig. 1.3

(d) With reference to Fig. 1.3,

(i) identify the level of organisation of the structure of ricin. [1]

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(ii) describe how the conformation of RTA (boxed section) is obtained from a linear polypeptide chain. [3]

(e) The RTB component of ricin has two binding sites specific for galactose. This enables RTB to bind to cell surface components that contain galactose. This is the first step of cellular intoxication. Identify a structure found on the mammalian cell surface that makes it a good target for ricin. [1]

(f) The RTA component of ricin is an enzyme that damages an essential component of the protein synthesis machinery – the ribosome. RTA catalyses the removal of one particular nitrogenous base (adenine) in a crucially important section of ribosomal RNA of the large subunit of eukaryotic ribosomes. This results in eventual cell death.

The two polypeptides – RTA and RTB are covalently linked by a disulfide bond between two cysteine residues. Reduction of ricin into its subunits, RTA and RTB, is catalysed by protein disulfide isomerase (PDI), an enzyme that resides in the lumen of the endoplasmic reticulum. PDI breaks the interchain disulfide bond.

Fig. 1.4 illustrates the reduction process catalysed by PDI.

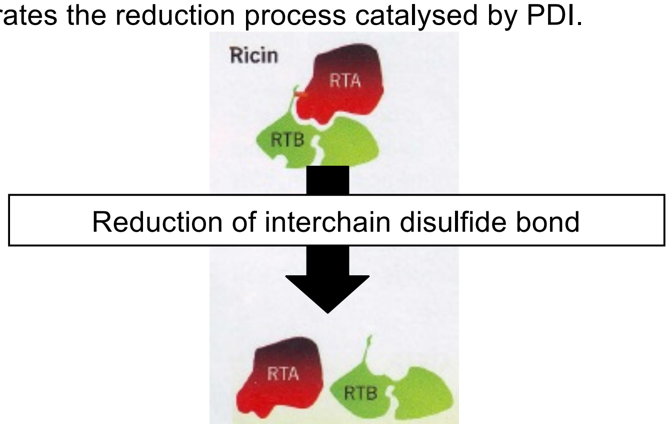

Fig. 1.4

(i) Explain why the reduction process shown in Fig. 1.4 is necessary before RTA can exhibit its catalytic activity. [2]

(ii) Suggest one way in which plant ribosomes in the seed cells are protected from the effect of ricin. [1]

[Total: 17]

2 **(a)** Fig. 2 is an electron micrograph of an organelle found in eukaryotic cells.

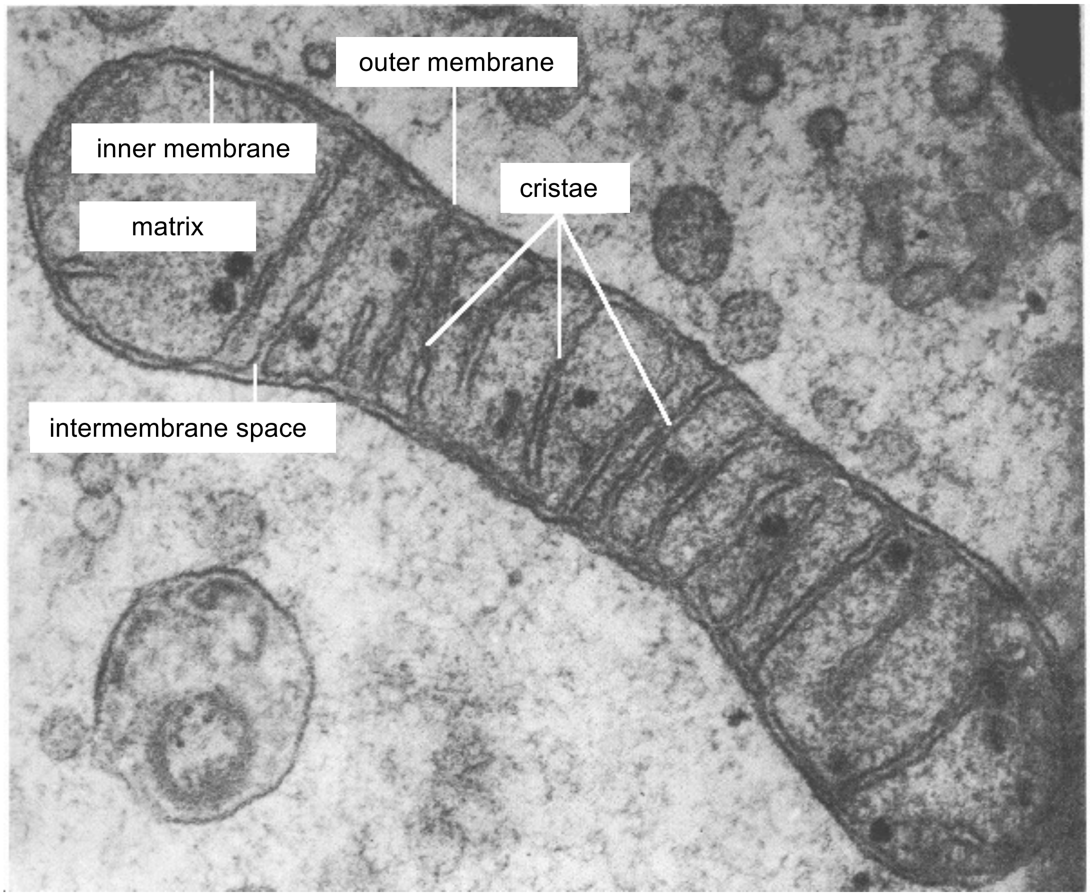

Fig. 2

State **two** similarities between the organelle above and a typical prokaryotic cell like *E. coli*. [2]

(b) **(i)** Mutants that are *lac Y* $^{-}$, meaning that the *lac Y* gene is inactivated, retain the capacity to synthesise β-galactosidase. However, even though the *lac I* gene encoding for the repressor protein is still intact, β-galactosidase can no longer be induced by adding lactose to the medium. Explain why this is so. [4]

(ii) Positive control of the *lac* operon results from breakdown of glucose, which alters the levels of cyclic adenosine monophosphate (cAMP) in the cell. cAMP binds to a catabolite activator protein (CAP), forming a CAP-cAMP complex which increases the affinity of RNA polymerase for the *lac* promoter. However, this mechanism will not cause upregulation (genetic regulation that leads to an increase in gene expression) of the *lac* operon in the absence of lactose. Explain why this is so. [2]

(iii) Suggest a reason how a mutation in the operon can result in a repressor that continually binds to the operator. [1]

(c) Why is any bacterial gene capable of being transferred during generalised transduction? [3]

[Total: 12]

3 The human papillomavirus (HPV) is a small non-enveloped virus that has a double-stranded circular DNA genome. Some HPVs have been well established as the main risk factor of cervical cancers.

The HPV genome is replicated in the nucleus of infected cells in the form of a circular DNA (episome). The HPV E2 gene encodes a transcription factor E2 that suppresses transcription of HPV E6 and E7, which are viral oncogenes. In certain instances during the replication cycle of HPV, the viral DNA can be linearised and integrated into chromosomal DNA of host cells. A breakpoint that disrupts the HPV E2 gene will prevent the synthesis of E2 proteins that normally regulate the transcription of E6 and E7.

Fig. 3.1 illustrates the integration of HPV DNA into the host chromosome.

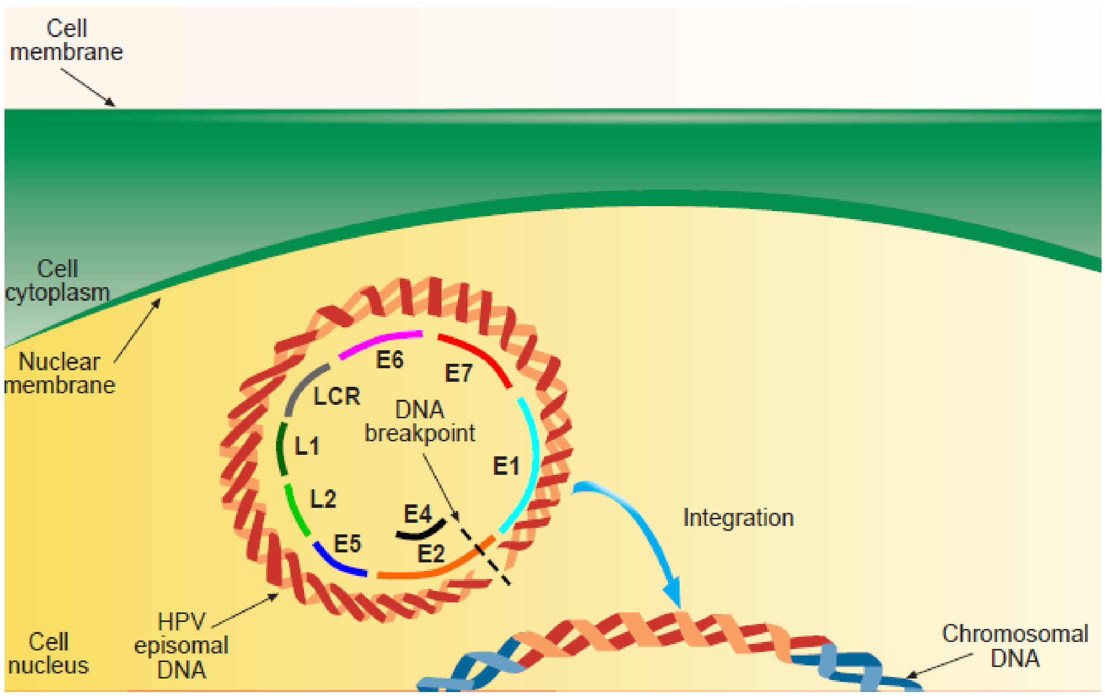

Fig. 3.1

Part of the HPV genome consisting of the long control region (LCR), E6 gene and E7 gene is integrated into the host DNA (Fig 3.2). The LCR contains a variety of regulatory sequences that will regulate viral replication and gene expression. The HPV E6 and E7 genes are consistently expressed, whereas the remaining HPV genes are often deleted or not transcribed after integration.

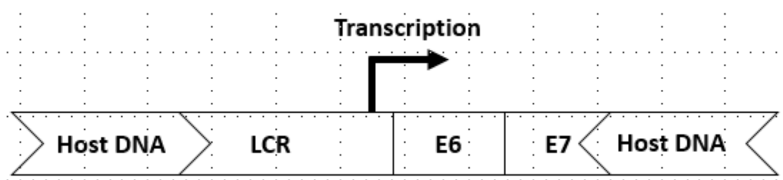

Fig. 3.2

(a) With reference to the information provided and Fig. 3.2, suggest how a dual-cistronic E6-E7 mRNA transcript is produced. [2]

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An increase in the synthesis of E6 and E7 oncoproteins is associated with the risk of development of carcinoma. Fig. 3.3 illustrates the HPV inactivation of two cellular tumour suppressor proteins, p53 and pRb.

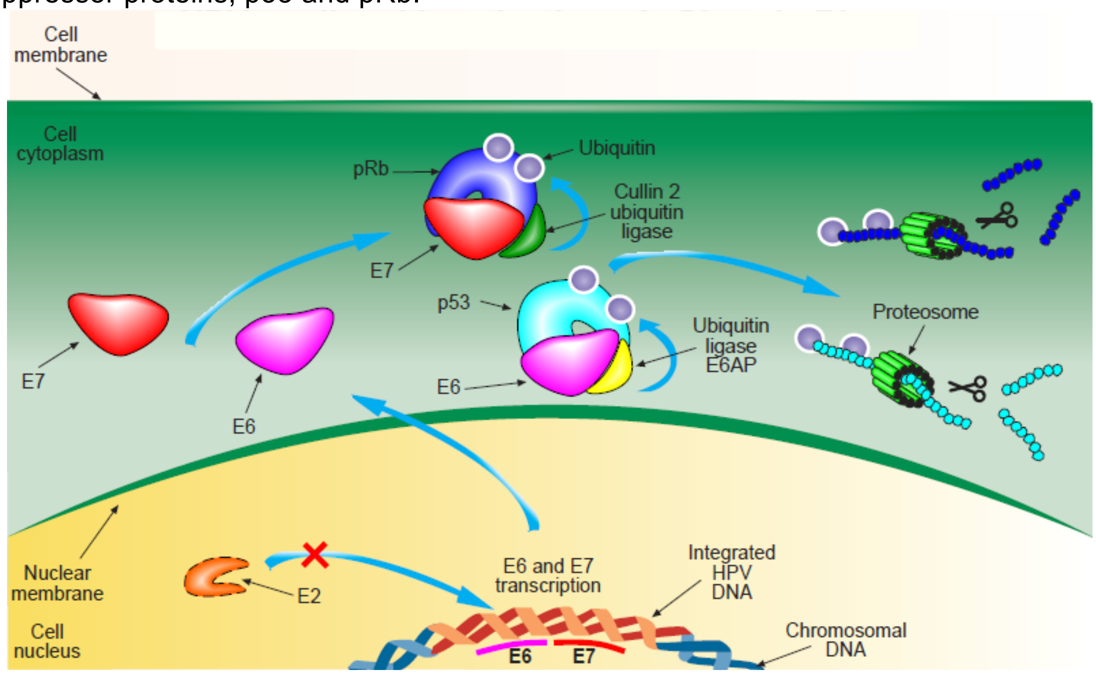

Fig. 3.3

(b) With reference to Fig. 3.3, describe how the absence of E2 and hence an increase in E6 and E7 protein synthesis, can result in the degradation of p53 and pRb. [2]

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(c) Explain why the degradation of p53 may lead to tumour formation. [3]

(d) The ends of eukaryotic chromosomes are protected by identical repeating lengths of DNA called telomeres, which prevent chromosomes from unravelling during cell division. Each time a cell divides, the length of the telomeres shortens and the ability of the cell to divide decreases.

Another function of the HPV E6 proteins is their ability to activate the expression of the catalytic subunit of telomerase (hTERT). Explain the relationship between high levels of telomerase expression and cancer. [4]

[Total: 11]

4 Wing colour in a butterfly species is determined by two alleles, one coding for red pigment and another coding for yellow pigment.

When true breeding yellow butterflies are mated with true breeding red butterflies, the wings of male progeny are always orange. Female progeny always have the same wing colour as their fathers.

(a) Using suitable symbols, draw **two** different genetic diagrams in the space below to fully explain the results obtained in the progeny. [5]

(b) Explain the mode of inheritance of wing colour in the butterfly species. [3]

(c) Observed results of the above genetic cross differ from the expected results.

Suggest **two** reasons why such a discrepancy occurs, referring only to events that occur after meiosis. [2]

[Total: 10]

5 Fig. 5 shows the steps involved during glycolysis.

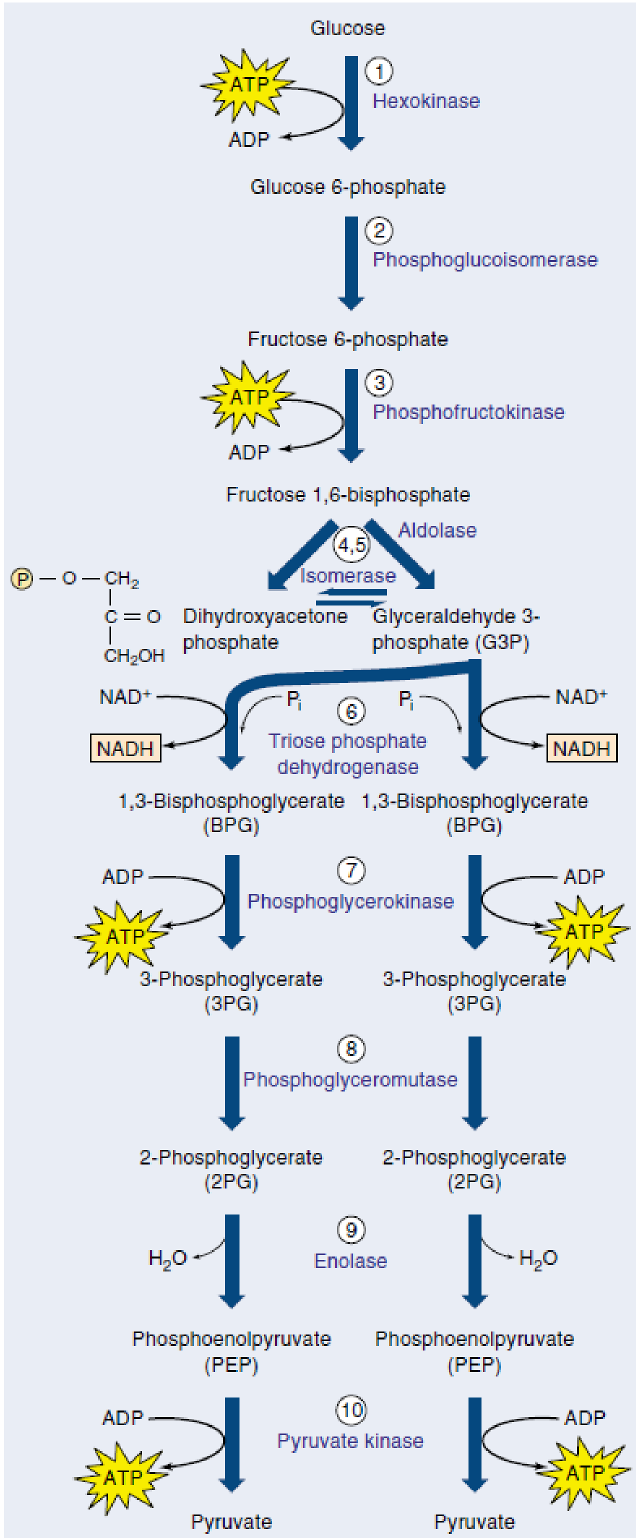

Fig. 5

(a) With reference to Fig. 5,

(i) explain the role of hexokinase in the first step of glycolysis. [2]

(ii) explain the role of glycolysis in aerobic respiration. [2]

(iii) describe how a named organism obtains energy for survival in the absence of oxygen from glycolysis. [2]

(b) Glyceraldehyde-3-phosphate is an intermediate in glycolysis which is also found in the Calvin cycle in plants.

Describe how glyceraldehyde-3-phosphate is formed in the Calvin cycle in plants. [4]

[Total: 10]

6 Cry toxins are produced by the soil bacterium *Bacillus thuringiensis* (Bt). By genetic engineering, Cry toxin gene can be incorporated into commercially important crops to specifically kill insect pests without affecting other organisms.

Cry toxins are believed to cause death in target insects by oligomerising into pore complexes in cell surface membrane of midgut epithelial cells.

(a) Explain how formation of pores in the cell surface membrane of midgut epithelial cells causes death in target insects. [1]

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Another mechanism of action of Cry toxin in midgut epithelial cells of target insects has been proposed. Fig. 6 shows the proposed magnesium ion-dependent signalling cascade. Single-pass membrane receptors that cross the membrane only once, such as aminopeptidase N (APN) receptors and cadherin-like receptors, are involved. Cry toxins eventually lead to destabilisation of the cytoskeleton and ion channels in the cell membrane causing cell death.

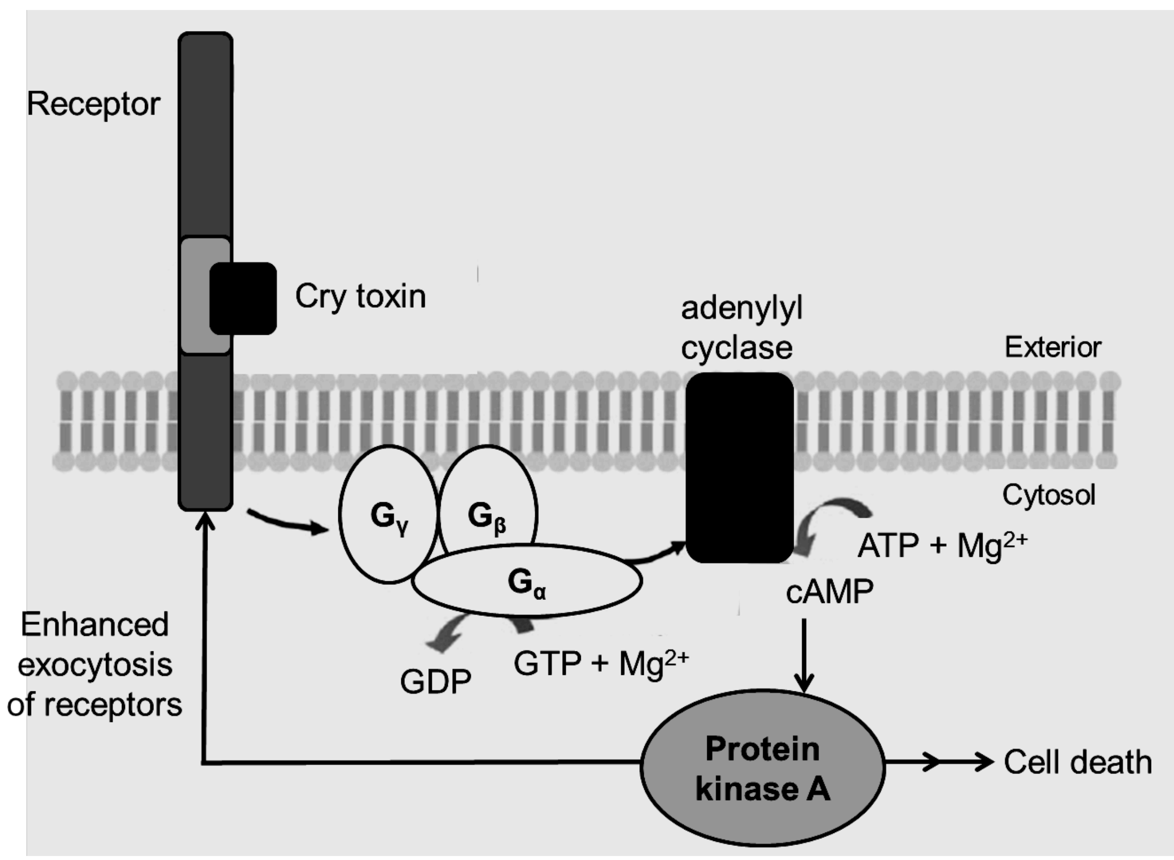


Fig. 6

(b) **(i)** With reference to Fig. 6, describe the proposed cell signalling cascade of Cry toxin. [4]

(ii) State **one** advantage and **one** disadvantage of a signalling cascade. [2]

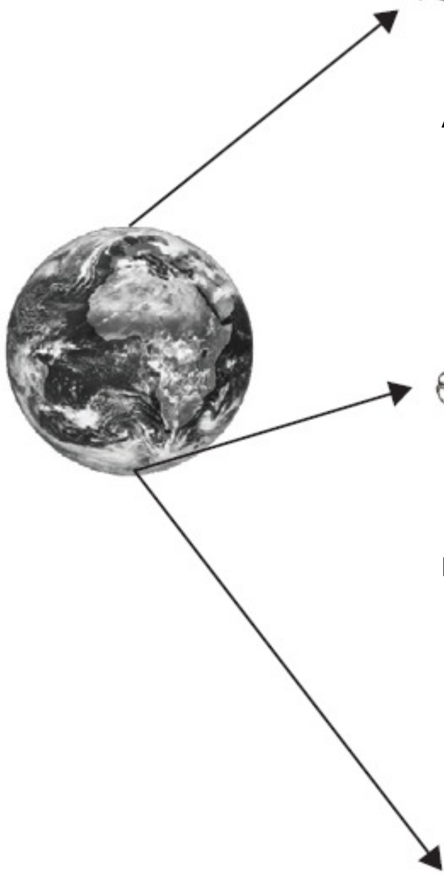
(iii) NF449 is an antagonist of G protein that prevents binding of GTP. Suggest how NF449 could affect the signalling pathway of Cry toxins. [2]

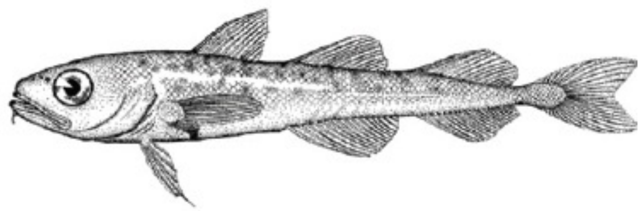
(c) Suggest why Cry toxins only kill target insects and not other organisms. [1]

[Total: 10]

7 Antifreezes are specialised proteins that prevent the organisms that have them from freezing in the polar oceans of the Arctic and the Antarctic. One of these proteins is the antifreeze glycoprotein (AFGP), which is produced in some fish species. The AFGP proteins protect the fish by lowering their bodies' freezing point temperature so that it is lower than that of the surrounding seawater. This prevents ice crystals from forming within the fishes' body tissues when fish in icy waters drink and feed, which would be lethal.

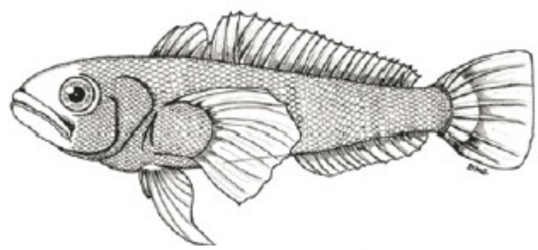
Fig. 7 shows three species of fish with AFGPs.





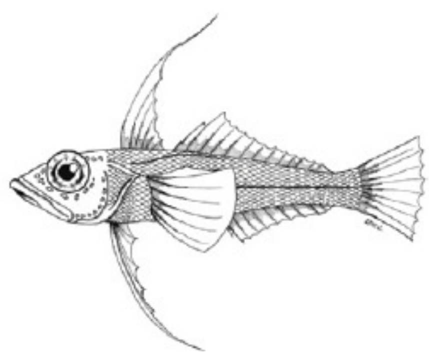
Arctic cod (Suborder: *Gadoilei*).

Lives at the water's surface and at depths below 900 metres in the Arctic.



Bald notothen (Suborder: *Notothenioidei*).

Lives on the underside of the surface ice layer in the Antarctic.



Threadfin pithead (Suborder: *Notothenioidei*).

Lives in the deep-sea habitats of the Antarctic (600 metres).

Fig. 7

A circumpolar current contributes to the thermal isolation and low temperatures of present-day Antarctica. This current formed about 30 million years ago when Antarctica separated from South America. Up until this time, the waters around Antarctica had been relatively warm and a wide diversity of fish species flourished.

With the separation of Antarctica from South America, water temperatures began to drop, reaching freezing levels about 10–15 million years ago. At this time, it is thought that the modern day notothenioids were represented by a single species in which the gene for AFGP may have arisen.

Today the seawater temperature around Antarctica seldom varies more than a few degrees above freezing. These waters are now dominated by the notothenioids, of which there are more than 100 species.

AFGP is also found in a species of arctic fish, the Arctic cod. This species is unrelated to the notothenioids, and its AFGP probably evolved about 4–5 million years ago during the glaciation of the Arctic seas. The Arctic cod and the notothenioids separated long before the antifreeze glycoproteins developed.

Researchers have found that the gene for AFGP in the Antarctic notothenioids is quite different in its origin and in its location within the genome from the gene expressing AFGP in the Arctic cod. However, their AFGPs are virtually identical and they both contain the same repeating sequence of three amino acids (threonine-alanine-alanine).

(a) Explain how natural selection could have brought about the evolution of antifreeze glycoprotein (AFGP) in the notothenioids in the seawater around Antarctica. [5]

(b) **(i)** What is the term used to describe the evolutionary pattern observed in the Antarctic notothenioids and the Arctic cod? [1]

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(ii) Explain how the evolutionary pattern you have identified in **(b)(i)** resulted in the evolution of virtually identical AFGPs in the Antarctic notothenioids and the Arctic cod. [4]

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[Total: 10]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where applicable.
Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

8 **(a)** Outline the roles of phospholipids, cholesterol and glycolipids in the cell surface membrane. [6]

(b) Explain how the molecular structure of haemoglobin is related to its function. [7]

(c) Describe the structural differences between collagen and cellulose. [7]

9 **(a)** Outline the ways in which mutations can occur. [6]

(b) Explain how gene mutation leads to cystic fibrosis. [7]

(c) Describe the role of mitosis in maintaining genetic stability. [7]